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LIGHT SOURCE APPARATUS AND PROJECTOR

Abstract

A light source apparatus includes a first light source configured to output first light; a first sensor configured to detect a first measured value corresponding to a temperature of the first light source; and a controller configured to control an output from the first light source based on the first measured value.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-028394, filed Feb. 28, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a light source apparatus and a projector.

2. Related Art

[0003] JP-A-2015-154163 discloses a projector including an external light quantity sensor that measures external illuminance and an internal light quantity sensor that measures the light quantity of the light from a light source via a light modulator, and that adjusts at least one of the light quantity of the light from the light source and the luminance of an image based on measured values provided from the two sensors.

[0004] JP-A-2015-154163 is an example of the related art.

[0005] In the related art described above, since it is necessary to provide two light quantity sensors in the projector, the number and the cost of the parts of the

SUMMARY

[0006] A light source apparatus according to an aspect of the present disclosure includes: a first light source configured to output first light; a first sensor configured to detect a first measured value corresponding to a temperature of the first light source; and a controller configured to control an output from the first light source based on the first measured value.

[0007] A projector according to another aspect of the present disclosure includes: the light source apparatus according to the aspect described above; a light modulator configured to modulate light output from the light source apparatus; and a projection system configured to project the light modulated by the light modulator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a block diagram diagrammatically showing the configuration of a light source apparatus according to a first embodiment.

[0009] FIG. **2** shows an example of a result of measurement of the output from a light source.

[0010] FIG. **3** is a flowchart showing a light source control process in the first embodiment.

[0011] FIG. **4** is a block diagram diagrammatically showing the configuration of a light source apparatus according to a second embodiment.

[0012] FIG. **5** is a flowchart showing a light source control process in the second embodiment.

[0013] FIG. **6** is a block diagram diagrammatically showing the configuration of a light source apparatus according to a third embodiment.

[0014] FIG. **7** is a flowchart showing a light source control process in the third embodiment.

[0015] FIG. **8** diagrammatically shows the configuration of a projector including a light source apparatus.

DESCRIPTION OF EMBODIMENTS

[0016] Embodiments of the present disclosure will be described below with reference to the drawings.

[0017] In the following drawings, elements may be drawn at different dimensional scales for clarity of the elements.

First Embodiment of Light Source Apparatus

[0018] FIG. **1** is a block diagram diagrammatically showing the configuration of a light source apparatus **10** according to a first embodiment. The light source apparatus **10** includes a first light source **11**B, a second light source **11**G, a third light source **11**R, a light combiner **12**, a first light source driver **13**B, a second light source driver **13**G, a third light source driver **13**R, a first sensor **14**B, a second sensor **14**G, a third sensor **14**R, and a controller **15**, as shown in FIG. **1**.

[0019] The first light source **11**B outputs first light BL. The first light source **11**B outputs blue light as the first light BL by way of example. For example, the first light source **11**B may include one or more blue laser diodes to output blue light as the first light BL. The first light source driver **13**B supplies the first light source **11**B with a first drive current. The output from the first light source **11**B is controlled by the first drive current.
[0020] The second light source **11**G outputs second light GL. The second light source **11**G outputs green light as the second light GL by way of example. For example, the second light source **11**G may include one or more green laser diodes to output green light as the second light GL. The second light source driver **13**G supplies the second light source **11**G with a second drive current. The output from the second light source **11**G is controlled by the second drive current.
[0021] The third light source **11**R outputs third light RL. The third light source **11**R outputs red light as the third light RL by way of example. For example, the third light source **11**R may include one or more red laser diodes to output red light as the third light RL. The third light source driver **13**R supplies the third light source **11**R with a third drive current. The output from the third light source **11**R is controlled by the third drive current.
[0022] The light combiner **12** combines the first light BL, the second light GL, and the third light RL with one another. The light combiner **12** outputs white light WL as a result of the combination of the first light BL, the second light GL, and the third light RL. That is, the light source apparatus **10** outputs the white light WL.
[0023] The first light source driver **13**B supplies the first light source **11**B with the first drive current based on a first control signal output from the controller **15**. The value of the first drive current supplied from the first light source driver **13**B to the first light source **11**B is controlled by the controller **15**.
[0024] The second light source driver **13**G supplies the second light source **11**G with the second drive current based on a second control signal output from the controller **15**. The value of the second drive current supplied from the second light source driver **13**G to the second light source **11**G is controlled by the controller **15**.
[0025] The third light source driver **13**R supplies the third light source **11**R with the third drive current based on a third control signal output from the controller **15**. The value of the third drive current supplied from the third light source driver **13**R to the third light source **11**R is controlled by the controller **15**.
[0026] The first sensor **14**B detects a first measured value corresponding to the temperature of the first light source **11**B. In the present embodiment, the first measured value is the temperature. The first measured value is not limited to the temperature, and may, for example, be the value of a voltage applied to the first light source **11**B. The first sensor **14**B outputs an electric signal indicating the first measured value to the controller **15**.
[0027] The second sensor **14**G detects a second measured value corresponding to the temperature of the second light source **11**G. In the present embodiment, the second measured value is the temperature. The second measured value is not limited to the temperature, and may, for example, be the value of a voltage applied to the second light source **11**G. The second sensor **14**G outputs an electric signal indicating the second measured value to the controller **15**.
[0028] The third sensor **14**R detects a third measured value corresponding to the temperature of the third light source **11**R. In the present embodiment, the third measured value is the temperature. The third measured value is not limited to the temperature, and may be the value of a voltage applied to the third light source **11**R. The third sensor **14**R outputs an electric signal indicating the third measured value to the controller **15**. For example, the first sensor **14**B, the second sensor **14**G, and the third sensor **14**R are each a thermistor. In general, a thermistor processes the output from the sensor in the form of voltage, and can therefore directly process the measured value when the measured value is a voltage.
[0029] The controller **15** separately controls the output from the first light source **11**B, the output

from the second light source **11**G, and the output from the third light source **11**R. As will be described later in detail, the controller **15** controls the output from the first light source **11**B based on the first measured value detected by the first sensor **14**B. The controller **15** further controls the output from the second light source **11**G based on the second measured value detected by the second sensor **14**G. Moreover, the controller **15** controls the output from the third light source **11**R based on the third measured value detected by the third sensor **14**R.

[0030] More specifically, the controller **15** controls the output from the first light source **11**B by controlling the first drive current, which is supplied to the first light source **11**B, based on the first measured value. The controller **15** further controls the output from the second light source **11**G by controlling the second drive current, which is supplied to the second light source **11**G, based on the second measured value. Moreover, the controller **15** controls the output from the third light source **11**R by controlling the third drive current, which is supplied to the third light source **11**R, based on the third measured value.

[0031] For example, the controller **15** includes one or more processors and one or more memories. For example, the processor is configured with a CPU (central processing unit). Some or all functions of the processor may be implemented by a circuit such as a digital signal processor (DSP), an application specific integrated circuit (ASIC), a programmable logic device (PLD), or a field programmable gate array (FPGA). The processor carries out various processes in parallel or sequentially.

[0032] The memory includes a nonvolatile memory that stores programs necessary for causing the processor to carry out the various processes, various setting data, and the like, and a volatile memory used as a temporary data saving destination when the processor carries out the various processes. For example, the nonvolatile memory is an electrically erasable programmable read-only memory (EEPROM) or a flash memory. The volatile memory is, for example, a random access memory (RAM).

[0033] Control parameters necessary for controlling the first light source **11**B, the second light source **11**G, and the third light source **11**R are stored in advance in the nonvolatile memory of the controller **15**. A method for acquiring the control parameters necessary for controlling the first light source **11**B, the second light source **11**G, and the third light source **11**R will be described below. The method for acquiring the control parameters includes first to fifth steps described below.

First Step: Set Measurement Conditions

[0034] In the first step, multiple different measurement conditions are set in accordance with the combination of the first measured value corresponding to the temperature of the first light source **11**B and the first drive current supplied to the first light source **11**B. In the following description, the first measured value set as a measurement condition may be referred to as a temperature condition value, and the value of the first drive current set as another measurement condition may be referred to as a current condition value.

Second Step: Measure Output from Light Source

[0035] In the second step, when the first light source **11**B is driven under each of the measurement conditions, the power of the first light BL output from the first light source **11**B and the values of the XYZ components in the CIE XYZ color system are measured. The CIE XYZ color system is an example of a first color system. In the following description, the value of measured power may be referred to as a power measured value, the value of a measured X component may be referred to as an X-component measured value, the value of a measured Y component may be referred to as a Y-component measured value, and the value of a measured Z component may be referred to as a Z-component measured value.

[0036] FIG. **2** shows an example of a result of the measurement of the output from a light source. For example, when the first light source **11**B is driven under a measurement condition that is the combination of a temperature condition value Tc.sub.B0 and a current condition value Ic.sub.B0, a power measured value Pm.sub.B0, an X-component measured value Xm.sub.B0, a Y-component measured value Ym.sub.B0, and a Z-component measured value Zm.sub.B0 are obtained, as shown in FIG. **2**.

[0037] Furthermore, for example, when the first light source **11**B is driven under a measurement condition that is the combination of a temperature condition value Tc.sub.B1 and a current condition value Ic.sub.B1, a power measured value Pm.sub.B1, an X-component measured value Xm.sub.B1, a Y-component measured value Ym.sub.B1, and an Z-component measured value Zm.sub.B1 are obtained, as shown in FIG. **2**.

[0038] Moreover, for example, when the first light source **11**B is driven under a measurement condition that is the combination of a temperature condition value TC.sub.Bn and a current condition value Ic.sub.Bn, a power measured value Pm.sub.Bn, an X-component measured value Xm.sub.Bn, a Y-component measured value Ym.sub.Bn, and a Z-component measured value Zm.sub.Bn are obtained, as shown in FIG. **2**.

Third Step: Calculate Output Fixing Coefficients

[0039] In the third step, output fixing coefficients are calculated based on the temperature condition value, the current condition value, and the power measured value. For example, the current condition value Ic.sub.B0 can be expressed by a polynomial having variables that are the temperature condition value TC.sub.B0 and the power measured value Pm.sub.B0, as indicated by Expression (1) below. Similarly, the current condition value Ic.sub.Bn can be expressed by a polynomial having variables that are the temperature condition value Tc.sub.Bn and the power measured value Pm.sub.Bn, as indicated by Expression (2) below. Note that other current condition values such as the current condition value Ic.sub.B1 can each be similarly expressed by a polynomial although not shown. That is, polynomials the number of which is equal to the number of measurement conditions are obtained. In Expressions (1) and (2), coefficients k.sub.B00 to k.sub.Bij are the output fixing coefficients.

[00001]

$$\left\{\begin{aligned} Ic_{B0} &= k_{B_{00}} \cdot Tc_{B0}^{0} \cdot Pm_{B0}^{0} + \cdots + k_{B_{ij}} \cdot Tc_{B0}^{i} \cdot Pm_{B0}^{j} \quad (1) \\ Ic_{Bn} &= k_{B_{00}} \cdot Tc_{Bn}^{0} \cdot Pm_{Bn}^{0} + \cdots + k_{B_{ij}} \cdot Tc_{Bn}^{i} \cdot Pm_{Bn}^{j} \quad (2) \end{aligned}\right.$$

[0040] When the polynomials described above are transformed into a determinant, Expression (3) below is obtained. The output fixing coefficients k.sub.B00 to k.sub.Bij are calculated by solving simultaneous equations based on Expression (3) below.

[00002]

$$\begin{pmatrix} Ic_{B0} \\ \vdots \\ Ic_{Bn} \end{pmatrix} = \begin{pmatrix} Tc_{B0}^{0} \cdot Pm_{B0}^{0} & \cdots & Tc_{B0}^{i} \cdot Pm_{B0}^{j} \\ \vdots & \ddots & \vdots \\ Tc_{xn}^{0} \cdot Pm_{B0}^{0} & \cdots & Tc_{Bn}^{0} \cdot Pm_{Bn}^{j} \end{pmatrix} \begin{pmatrix} k_{B_{00}} \\ \vdots \\ k_{B_{ij}} \end{pmatrix} \quad (3)$$

[0041] Note that the relationship among n, i, and j in Expressions (1), (2), and (3) is expressed by Expression (4) below.

[00003] $n = (i + 1)(j + 1)$ (4)

Fourth Step: Acquire Standard Values of XYZ Components

[0042] In the fourth step, standard values of the XYZ components are acquired. For example, any one of the X-component measured values is acquired as a standard value X.sub.B0 of the X component. Furthermore, any one of the Y-component measured values is acquired as a standard value Y.sub.B0 of the Y component. Moreover, any one of the Z-component measured values is acquired as a standard value Z.sub.B0 of the Z component.

Fifth Step: Calculate Wavelength Shift Correction Coefficients

[0043] In the fifth step, wavelength shift correction coefficients are calculated. First, spectra of the first light BL output from the first light source **11**B are measured, and the measured spectra are so normalized that the sum of the spectra becomes one. The value of the X component is then calculated from the normalized spectra. The proportion of the X component is then calculated by dividing the calculated value of the X component by the standard value X.sub.B0 of the X component. The proportion of the X component is then divided by the temperature condition value used in the spectrum measurement to calculate a wavelength shift correction coefficient K.sub.BX of the X component.

[0044] The value of the Y component is then calculated from the normalized spectra. The proportion of the Y component is then calculated by dividing the calculated value of the Y component by the standard value Y.sub.B0 of the Y component. The proportion of the Y component is then divided by the temperature condition value used in the spectrum measurement to calculate a wavelength shift correction coefficient K.sub.BY of the Y component.

[0045] The value of the Z component is then calculated from the normalized spectra. The proportion of the Z component is then calculated by dividing the calculated value of the Z component by the standard value Z.sub.B0 of the Z component. The proportion of the Z component is then divided by the temperature condition value used in the spectrum measurement to calculate a wavelength shift correction coefficient K.sub.BZ of the Z component.

[0046] As described the above, output fixing coefficients k.sub.B00 to k.sub.Bij, the standard value X.sub.B0 of the X component, the standard value Y.sub.B0 of the Y component, the standard value Z.sub.B0 of the Z component, the wavelength shift correction coefficient K.sub.BX of the X component, the wavelength shift correction coefficient K.sub.BY of the Y component, and the wavelength shift correction coefficient K.sub.BZ of the Z component are obtained as the control parameters necessary for controlling the first light source **11**B by executing the first to fifth steps described above for the first light source **11**B.

[0047] The output fixing coefficients k.sub.B00 to k.sub.Bij are an example of a first coefficient set in advance in correspondence with the first light source **11**B. The standard value X.sub.B0 of the X component, the standard value Y.sub.B0 of the Y component, and the standard value Z.sub.B0 of the Z component are examples of a first standard value expressed in the first color system and set in advance in correspondence with the first light source **11**B. The wavelength shift correction coefficient K.sub.BX of the X component, the wavelength shift correction coefficient K.sub.BY of the Y component, and the wavelength shift correction coefficient K.sub.BZ of the Z component are examples of a first correction coefficient set in advance in correspondence with the first light source **11**B.

[0048] Similarly, output fixing coefficients k.sub.G00 to k.sub.Gij, a standard value X.sub.G0 of the X component, a standard value Y.sub.G0 of the Y component, a standard value Z.sub.G0 of the Z component, a wavelength shift correction coefficient K.sub.GX of the X component, a wavelength shift correction coefficient K.sub.GY of the Y component, and a wavelength shift correction coefficient K.sub.GZ of the Z component are obtained as the control parameters necessary for controlling the second light source **11**G by executing the first to fifth steps described above for the second light source **11**G.

[0049] The output fixing coefficients k.sub.G00 to K.sub.Gij are an example of a second coefficient set in advance in correspondence with the second light source **11**G. The standard value X.sub.G0 of the X component, the standard value Y.sub.G0 of the Y component, and the standard value Z.sub.G0 of the Z component are examples of a second standard value expressed in the first color system and set in advance in correspondence with the second light source **11**G. The wavelength shift correction coefficient K.sub.GX of the X component, the wavelength shift correction coefficient K.sub.GY of the Y component, and the wavelength shift correction coefficient K.sub.GZ of the Z component are examples of a second correction coefficient set in advance in correspondence with the second light source **11**G.

[0050] Similarly, output fixing coefficients k.sub.R00 to K.sub.Rij, a standard value X.sub.R0 of the X component, a standard value Y.sub.R0 of the Y component, a standard value Z.sub.R0 of the Z component, a wavelength shift correction coefficient K.sub.RX of the X component, a wavelength shift correction coefficient K.sub.RY of the Y component, and a wavelength shift correction coefficient K.sub.RZ of the Z component are obtained as the control parameters

necessary for controlling the third light source **11**R by executing the first to fifth steps described above for the third light source **11**R.
[0051] The output fixing coefficients k.sub.R00 to k.sub.Rij are an example of a third coefficient set in advance in correspondence with the third light source **11**R. The standard value X.sub.R0 of the X component, the standard value Y.sub.R0 of the Y component, and the standard value Z.sub.R0 of the Z component are examples of a third standard value expressed in the first color system and set in advance in correspondence with the third light source **11**R. The wavelength shift correction coefficient K.sub.RX of the X component, the wavelength shift correction coefficient K.sub.RY of the Y component, and the wavelength shift correction coefficient K.sub.RZ of the Z component are examples of a third correction coefficient set in advance in correspondence with the third light source **11**R.
[0052] The method for acquiring the control parameters stored in advance in the nonvolatile memory of the controller **15** has been described above. A light source control process carried out by the controller **15** will be described below with reference to FIG. **3**. FIG. **3** is a flowchart showing the light source control process carried out by the controller **15**.
[0053] The controller **15** first sets a target value of the Y component and target values of the color coordinate xy (step S**1**), as shown in FIG. **3**. The target value of the Y component and the target values of the color coordinate xy are examples of a target value expressed in the first color system.
[0054] The controller **15** subsequently measures values that change in accordance with the temperatures of the first light source **11**B, the second light source **11**G, and the third light source **11**R (step S**2**). Specifically, the controller **15** acquires a first measured value Tm.sub.B corresponding to the temperature of the first light source **11**B based on an output signal from the first sensor **14**B. The controller **15** further acquires a second measured value Tm.sub.G corresponding to the temperature of the second light source **11**G based on the output signal from the second sensor **14**G. The controller **15** still further acquires a third measured value Tm.sub.R corresponding to the temperature of the third light source **11**R based on the output signal from the third sensor **14**R.
[0055] The controller **15** subsequently carries out a wavelength shift correction process for the first light source **11**B (step S**3**). Specifically, the controller **15** reads the standard value X.sub.B0 of the X component and the wavelength shift correction coefficient K.sub.BX of the X component out of the control parameters relating to the first light source **11**B from the memory. The controller **15** then corrects the standard value X.sub.B0 of the X component based on the first measured value Tm.sub.B and the wavelength shift correction coefficient K.sub.BX of the X component. Specifically, the controller **15** calculates a reference value X.sub.B1 of the X component relating to the first light source **11**B by substituting the standard value X.sub.B0 of the X component, the wavelength shift correction coefficient K.sub.BX of the X component, and the first measured value Tm.sub.B into Expression (5) below.
[0056] The controller **15** further reads the standard value Y.sub.B0 of the Y component and the wavelength shift correction coefficient K.sub.BY of the Y component out of the control parameters relating to the first light source **11**B from the memory. The controller **15** then corrects the standard value Y.sub.B0 of the Y component based on the first measured value Tm.sub.B and the wavelength shift correction coefficient K.sub.BY of the Y component. Specifically, the controller **15** calculates a reference value Y.sub.B1 of the Y component relating to the first light source **11**B by substituting the standard value Y.sub.B0 of the Y component, the wavelength shift correction coefficient K.sub.BY of the Y component, and the first measured value Tm.sub.B into Expression (6) below.
[0057] The controller **15** still further reads the standard value Z.sub.B0 of the Z component and the wavelength shift correction coefficient K.sub.BZ of the Z component out of the control parameters relating to the first light source **11**B from the memory. The controller **15** then corrects the standard value Z.sub.B0 of the Z component based on the first measured value Tm.sub.B and the

wavelength shift correction coefficient K.sub.BZ of the Z component. Specifically, the controller **15** calculates a reference value Z.sub.B1 of the Z component relating to the first light source **11**B by substituting the standard value Z.sub.B0 of the Z component, the wavelength shift correction coefficient K.sub.BZ of the Z component, and the first measured value Tm.sub.B into Expression (7) below.

[00004]{ $$\begin{aligned} X_{B1} &= X_{B0} \text{ .Math. } K_{BX} \text{ .Math. } \mathrm{Tm}_B \quad (5) \\ Y_{B1} &= Y_{B0} \text{ .Math. } K_{BY} \text{ .Math. } \mathrm{Tm}_B \quad (6) \\ Z_{B1} &= Z_{B0} \text{ .Math. } K_{BZ} \text{ .Math. } \mathrm{Tm}_B \quad (7) \end{aligned}$$

[0058] The controller **15** subsequently carries out the wavelength shift correction process for the second light source **11**G (step S**4**). Specifically, the controller **15** reads the standard value X.sub.G0 of the X component and the wavelength shift correction coefficient K.sub.GX of the X component out of the control parameters relating to the second light source **11**G from the memory. The controller **15** then corrects the standard value X.sub.G0 of the X component based on the second measured value Tm.sub.G and the wavelength shift correction coefficient K.sub.GX of the X component. Specifically, the controller **15** calculates a reference value X.sub.G1 of the X component relating to the second light source **11**G by substituting the standard value X.sub.G0 of the X component, the wavelength shift correction coefficient K.sub.GX of the X component, and the second measured value Tm.sub.G into Expression (8) below.

[0059] The controller **15** further reads the standard value Y.sub.G0 of the Y component and the wavelength shift correction coefficient K.sub.GY of the Y component out of the control parameters relating to the second light source **11**G from the memory. The controller **15** then corrects the standard value Y.sub.G0 of the Y component based on the second measured value Tm.sub.G and the wavelength shift correction coefficient K.sub.GY of the Y component. Specifically, the controller **15** calculates a reference value Y.sub.G1 of the Y component relating to the second light source **11**G by substituting the standard value Y.sub.G0 of the Y component, the wavelength shift correction coefficient K.sub.GY of the Y component, and the second measured value Tm.sub.G into Expression (9) below.

[0060] The controller **15** still further reads the standard value Z.sub.G0 of the Z component and the wavelength shift correction coefficient K.sub.GZ of the Z component out of the control parameters relating to the second light source **11**G from the memory. The controller **15** then corrects the standard value Z.sub.G0 of the Z component based on the second measured value Tm.sub.G and the wavelength shift correction coefficient K.sub.GZ of the Z component. Specifically, the controller **15** calculates a reference value Z.sub.G1 of the Z component relating to the second light source **11**G by substituting the standard value Z.sub.G0 of the Z component, the wavelength shift correction coefficient K.sub.GZ of the component, and the second measured value Tm.sub.G into Expression (10) below.

[00005]{ $$\begin{aligned} X_{G1} &= X_{G0} \text{ .Math. } K_{\mathrm{GX}} \text{ .Math. } \mathrm{Tm}_G \quad (8) \\ Y_{G1} &= Y_{G0} \text{ .Math. } K_{\mathrm{GY}} \text{ .Math. } \mathrm{Tm}_G \quad (9) \\ Z_{G1} &= Z_{G0} \text{ .Math. } K_{\mathrm{GZ}} \text{ .Math. } \mathrm{Tm}_G \quad (10) \end{aligned}$$

[0061] The controller **15** subsequently carries out the wavelength shift correction process for the third light source **11**R (step S**5**). Specifically, the controller **15** reads the standard value X.sub.R0 of the X component and the wavelength shift correction coefficient K.sub.RX of the X component out of the control parameters relating to the third light source **11**R from the memory. The controller **15** then corrects the standard value X.sub.R0 of the X component based on the third measured value Tm.sub.R and the wavelength shift correction coefficient K.sub.RX of the X component. Specifically, the controller **15** calculates a reference value X.sub.R1 of the X component relating to the third light source **11**R by substituting the standard value X.sub.R0 of the X component, the wavelength shift correction coefficient K.sub.RX of the X component, and the third measured value Tm.sub.R into Expression (11) below.

[0062] The controller **15** further reads the standard value Y.sub.R0 of the Y component and the wavelength shift correction coefficient K.sub.RY of the Y component out of the control parameters relating to the third light source **11**R from the memory. The controller **15** then corrects the standard value Y.sub.R0 of the Y component based on the third measured value Tm.sub.R and the wavelength shift correction coefficient K.sub.RY of the Y component. Specifically, the controller **15** calculates a reference value Y.sub.R1 of the Y component relating to the third light source **11**R by substituting the standard value Y.sub.R0 of the Y component, the wavelength shift correction coefficient K.sub.RY of the component, and the third measured value Tm.sub.R into Expression (12) below.

[0063] The controller **15** still further reads the standard value Z.sub.R0 of the Z component and the wavelength shift correction coefficient K.sub.RZ of the Z component out of the control parameters relating to the third light source **11**R from the memory. The controller **15** then corrects the standard value Z.sub.R0 of the Z component based on the third measured value Tm.sub.R and the wavelength shift correction coefficient K.sub.RZ of the Z component. Specifically, the controller **15** calculates a reference value Z.sub.R1 of the Z component relating to the third light source **11**R by substituting the standard value Z.sub.R0 of the Z component, the wavelength shift correction coefficient K.sub.RZ of the Z component, and the third measured value Tm.sub.R into Expression (13) below.

[00006]{
$$X_{R1} = X_{R0} \text{ .Math. } K_{RX} \text{ .Math. } \mathrm{Tm}_R \quad (11)$$
$$Y_{R1} = Y_{R0} \text{ .Math. } K_{RY} \text{ .Math. } \mathrm{Tm}_R \quad (12)$$
$$Z_{R1} = Z_{R0} \text{ .Math. } K_{RZ} \text{ .Math. } \mathrm{Tm}_R \quad (13)$$

[0064] The controller **15** subsequently carries out a color matching process (step S**6**). Specifically, the controller **15** calculates a first power setting value Po.sub.B for the first light source **11**B, a second power setting value Po.sub.G for the second light source **11**G, and a third power setting value Po.sub.R for the third light source **11**R. A target value of the X component is expressed by Expression (14) below. A target value of the Y component is expressed by Expression (15) below. A target value of the Z component is expressed by Expression (16) below.

[00007]{
$$X = \tfrac{Y}{y}x = \mathrm{Po}_R \text{ .Math. } X_{R1} + Po_G \text{ .Math. } X_{G1} + Po_B \text{ .Math. } X_{B1} \quad (14)$$
$$Y = Y = Po_R \text{ .Math. } Y_{R1} + Po_G \text{ .Math. } Y_{G1} + Po_B \text{ .Math. } Y_{B1} \quad (15)$$
$$Z = \tfrac{Y}{y}(1 - x - y) = Po_R \text{ .Math. } Z_{R1} + Po_G \text{ .Math. } Z_{G1} + Po_B \text{ .Math. } Z_{B1} \quad (16)$$

[0065] When Expressions (14), (15), and (16) described above are transformed into a determinant, Expression (17) below is obtained. The controller **15** solves simultaneous equations based on Expression (17) below to calculate the first power setting value Po.sub.B for the first light source **11**B, the second power setting value Po.sub.G for the second light source **11**G, and the third power setting value Po.sub.R for the third light source **11**R.

[00008]
$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} \tfrac{Y}{y}x \\ Y \\ \tfrac{Y}{y}(1 - x - y) \end{pmatrix} = \begin{pmatrix} X_{R1} & X_{G1} & X_{B1} \\ Y_{R1} & Y_{G1} & Y_{B1} \\ Z_{R1} & Z_{G1} & Z_{B1} \end{pmatrix} \begin{pmatrix} \mathrm{Po}_R \\ \mathrm{Po}_G \\ \mathrm{Po}_B \end{pmatrix} \quad (17)$$

[0066] The controller **15** subsequently carries out an output fixing process for the first light source **11**B (step S**7**). Specifically, the controller **15** reads the output fixing coefficients k.sub.B00 to k.sub.Bij out of the control parameters relating to the first light source **11**B from the memory. The controller **15** then calculates a first drive current setting value If.sub.B based on the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B. Specifically, the controller **15** calculates the first drive current setting value If.sub.B by substituting the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B into Expression (18) below.

[00009] $\text{If}_B = k_{B_{00}}$ .Math. Tm_B^0 .Math. $\text{Po}_B^0 +$ .Math. $+ k_{B_{ij}}$ .Math. Tm_B^i .Math. Po_B^j (18)

[0067] The controller **15** subsequently carries out the output fixing process for the second light source **11**G (step S**8**). Specifically, the controller **15** reads the output fixing coefficients k.sub.G00 to k.sub.Gij out of the control parameters relating to the second light source **11**G from the memory. The controller **15** then calculates a second drive current setting value If.sub.G based on the output fixing coefficients k.sub.G00 to K.sub.Gij, the second measured value Tm.sub.G, and the second power setting value Po.sub.G. Specifically, the controller **15** calculates the second drive current setting value If.sub.G by substituting the output fixing coefficients k.sub.G00 to k.sub.Gij, the second measured value Tm.sub.G, and the second power setting value Po.sub.G into Expression (19) below.

[00010] $\text{If}_G = k_{G_{00}}$ .Math. Tm_G^0 .Math. $\text{Po}_G^0 +$ .Math. $+ k_{G_{ij}}$ .Math. Tm_G^i .Math. Po_G^j (19)

[0068] The controller **15** subsequently carries out the output fixing process for the third light source **11**R (step S**9**). Specifically, the controller **15** reads the output fixing coefficients k.sub.R00 to k.sub.Rij out of the control parameters relating to the third light source **11**R from the memory. The controller **15** then calculates a third drive current setting value If.sub.R based on the output fixing coefficients k.sub.R00 to k.sub.Rij, the third measured value Tm.sub.R, and the third power setting value Po.sub.R. Specifically, the controller **15** calculates the third drive current setting value If.sub.R by substituting the output fixing coefficients k.sub.R00 to k.sub.Rij, the third measured value Tm.sub.R, and the third power setting value Po.sub.R into Expression (20) below.

[00011] $\text{If}_R = k_{R_{00}}$ .Math. Tm_R^0 .Math. $\text{Po}_R^0 +$ .Math. $+ k_{R_{ij}}$ .Math. Tm_R^i .Math. Po_R^j (20)

[0069] The controller **15** subsequently controls the drive current supplied to each of the light sources (step S**10**). Specifically, the controller **15** controls the first light source driver **13**B in such a way that the value of the first drive current supplied from the first light source driver **13**B to the first light source **11**B becomes the first drive current setting value If.sub.B. The controller **15** further controls the second light source driver **13**G in such a way that the value of the second drive current supplied from the second light source driver **13**G to the second light source **11**G becomes the second drive current setting value If.sub.G. The controller **15** still further controls the third light source driver **13**R in such a way that the value of the third drive current supplied from the third light source driver **13**R to the third light source **11**R becomes the third drive current setting value If.sub.R.

[0070] After the process in step S**10** ends, the controller **15** returns to step S**2** and repeatedly carries out the processes from steps S**2** to S**10** in a predetermined cycle.

Advantages of first embodiment

[0071] As described above, the light source apparatus **10** according to the first embodiment includes the first light source **11**B, which outputs the first light BL, the second light source **11**G, which outputs the second light GL, the third light source **11**R, which outputs the third light RL, the first sensor **14**B, which detects the first measured value corresponding to the temperature of the first light source **11**B, the second sensor **14**G, which detects the second measured value corresponding to the temperature of the second light source **11**G, the third sensor **14**R, which detects the third measured value corresponding to the temperature of the third light source **11**R, and the controller **15**. The controller **15** controls the output from the first light source **11**B based on the first measured value, controls the output from the second light source **11**G based on the second measured value, and controls the output from the third light source **11**R based on the third measured value.

[0072] According to the light source apparatus **10** described above, since the outputs from the first light source **11**B, the second light source **11**G, and the third light source **11**R are controlled by using three sensors less expensive than the light quantity sensors used in the related art, the cost of the parts of the light source apparatus **10** can be reduced. Since the outputs from the first light source **11**B, the second light source **11**G, and the third light source **11**R are separately controlled based on the temperatures of the first light source **11**B, the second light source **11**G, and the third light source **11**R, respectively, the white light WL output from the light source apparatus **10** can be accurately controlled.

[0073] In the light source apparatus **10** according to the first embodiment, the controller **15** controls the output from the first light source **11**B by controlling the first drive current, which is supplied to the first light source **11**B, based on the first measured value. The controller **15** further controls the output from the second light source **11**G by controlling the second drive current, which is supplied to the second light source **11**G, based on the second measured value. The controller **15** still further controls the output from the third light source **11**R by controlling the third drive current, which is supplied to the third light source **11**R, based on the third measured value.

[0074] According to the light source apparatus **10** described above, the output from the first light source **11**B can be readily controlled by controlling the first drive current based on the first measured value. The output from the second light source **11**G can also be readily controlled by controlling the second drive current based on the second measured value. Furthermore, the output from the third light source **11**R can be readily controlled by controlling the third drive current based on the third measured value.

[0075] In the light source apparatus **10** according to the first embodiment, the controller **15** calculates the first power setting value Po.sub.B for the first light source **11**B, the second power setting value Po.sub.G for the second light source **11**G, and the third power setting value Po.sub.R for the third light source **11**R based on the target values (Yxy) expressed in the XYZ color system, the first standard value (X.sub.B0, Y.sub.B0, Z.sub.B0) expressed in the XYZ color system and set in advance in correspondence with the first light source **11**B, the second standard value (X.sub.G0, Y.sub.G0, Z.sub.G0) expressed in the XYZ color system and set in advance in correspondence with the second light source **11**G, and the third standard value (X.sub.R0, Y.sub.R0, Z.sub.R0) expressed in the XYZ color system and set in advance in correspondence with the third light source **11**R. The controller **15** calculates the first drive current setting value If.sub.B based on the first measured value, the first coefficient (k.sub.B00 to k.sub.Bij) set in advance in correspondence with the first light source **11**B, and the first power setting value Po.sub.B. The controller **15** calculates the second drive current setting value If.sub.G based on the second measured value, the second coefficient (k.sub.G00 to k.sub.Gij) set in advance in correspondence with the second light source **11**G, and the second power setting value Po.sub.G. The controller **15** calculates the third drive current setting value If.sub.R based on the third measured value, the third coefficient (k.sub.R00 to k.sub.Rij) set in advance in correspondence with the third light source **11**R, and the third power setting value Po.sub.R.

[0076] According to the light source apparatus **10** described above, drive current setting values that can achieve the target values can be accurately calculated in accordance with changes in the temperatures of the light sources.

[0077] In the light source apparatus **10** according to the first embodiment, before calculating the first power setting value Po.sub.B, the second power setting value Po.sub.G, and the third power setting value Po.sub.R, the controller **15** corrects the first standard value based on the first measured value and the first correction coefficient set in advance in correspondence with the first light source **11**B, corrects the second standard value based on the second measured value and the second correction coefficient set in advance in correspondence with the second light source **11**G, and corrects the third standard value based on the third measured value and the third correction coefficient set in advance in correspondence with the third light source **11**R.

[0078] According to the light source apparatus **10** described above, even when wavelength shifts occur due to changes in the temperatures of the light sources, the first, second, and third standard values are corrected in terms of temperature, so that the drive current setting values can be accurately calculated in accordance with the changes in the temperatures of the light sources.

Second Embodiment of Light Source Apparatus

[0079] FIG. **4** is a block diagram diagrammatically showing the configuration of a light source apparatus **20** according to a second embodiment. The light source apparatus **20** includes a first light source **21**, a second light source **22**, a phosphor **23**, a light combiner **24**, a first light source driver **25**, a second light source driver **26**, a first sensor **27**, a second sensor **28**, and a controller **29**, as shown in FIG. **4**.

[0080] The first light source **21** outputs first light BL**1**. The first light source **21** outputs blue light as the first light BL**1** by way of example. For example, the first light source **21** may include one or more blue laser diodes to output blue light as the first light BL**1**. The first light source driver **25** supplies the first light source **21** with the first drive current. The output from the first light source **21** is controlled by the first drive current.

[0081] The second light source **22** outputs second light BL**2**. The second light source **22** outputs blue light as the second light BL**2** by way of example. For example, the second light source **22** may include one or more blue laser diodes to output blue light as the second light BL**2**. The second light source driver **26** supplies the second light source **22** with the second drive current. The output from the second light source **22** is controlled by the second drive current.

[0082] The phosphor **23** converts the second light BL**2** output from the second light source **22** into fourth light YL, which is yellow light. The light combiner **24** combines the first light BL**1**, which is blue light, and the fourth light YL, which is yellow light. The light combiner **12** outputs white light WL as a result of the combination of the first light BL**1** and the fourth light YL. That is, the light source apparatus **20** outputs the white light WL.

[0083] The first light source driver **25** supplies the first light source **21** with the first drive current based on a first control signal output from the controller **29**. The value of the first drive current supplied from the first light source driver **25** to the first light source **21** is controlled by the controller **29**.

[0084] The second light source driver **26** supplies the second light source **22** with the second drive current based on a second control signal output from the controller **29**. The value of the second drive current supplied from the second light source driver **26** to the second light source **22** is controlled by the controller **29**.

[0085] The first sensor **27** detects a first measured value corresponding to the temperature of the first light source **21**. The first sensor **27** outputs an electric signal indicating the first measured value to the controller **29**. The second sensor **28** detects a second measured value corresponding to the temperature of the second light source **22**. The second sensor **28** outputs an electric signal indicating the second measured value to the controller **29**. For example, the first sensor **27** and the second sensor **28** are each a thermistor.

[0086] The controller **29** separately controls the output from the first light source **21** and the output from the second light source **22**. As will be described later in detail, the controller **29** controls the output from the first light source **21** based on the first measured value detected by the first sensor **27**. The controller **29** controls the output from the second light source **22** based on the second measured value detected by the second sensor **28**.

[0087] More specifically, the controller **29** controls the output from the first light source **21** by controlling the first drive current, which is supplied to the first light source **21**, based on the first measured value. The controller **29** further controls the output from the second light source **22** by controlling the second drive current, which is supplied to the second light source **22**, based on the second measured value.

[0088] For example, the controller **29** includes one or more processors and one or more memories, and control parameters necessary for controlling the first light source **21** and the second light source **22** are stored in advance in a nonvolatile memory of the controller **29**, as in the controller **15** in the first embodiment. The method for acquiring the control parameters is the same as that in the first embodiment, and is therefore not repeatedly described in the second embodiment.

[0089] As the control parameters necessary for controlling the first light source **21**, output fixing coefficients k.sub.B00 to k.sub.Bij, a standard value X.sub.B0 of the X component, a standard value Y.sub.B0 of the Y component, a standard value Z.sub.B0 of the Z component, a wavelength shift correction coefficient K.sub.BX of the X component, a wavelength shift correction coefficient K.sub.BY of the Y component, and a wavelength shift correction coefficient K.sub.BZ of the Z component are stored in the memory.
[0090] As the control parameters necessary for controlling the second light source **22**, output fixing coefficients k.sub.Y00 to k.sub.Yij, a standard value X.sub.Y0 of the X component, a standard value Y.sub.Y0 of the Y component, a standard value Z.sub.Y0 of the Z component, a wavelength shift correction coefficient K.sub.YX of the X component, a wavelength shift correction coefficient K.sub.YY of the Y component, and a wavelength shift correction coefficient K.sub.YZ of the Z component are stored in the memory.
[0091] A light source control process carried out by the controller **29** will be described below with reference to FIG. **5**. FIG. **5** is a flowchart showing the light source control process carried out by the controller **29**.
[0092] The controller **29** first sets a target value of the Y component and target values of the color coordinate xy (step S**11**), as shown in FIG. **5**. The target value of the Y component and the target values of the color coordinate xy are examples of a target value expressed in the first color system.
[0093] The controller **29** subsequently measures values that change in accordance with the temperatures of the first light source **21** and the second light source **22** (step S**12**). Specifically, the controller **29** acquires a first measured value Tm.sub.B corresponding to the temperature of the first light source **21** based on the output signal from the first sensor **27**. The controller **29** further acquires a second measured value Tm.sub.Y corresponding to the temperature of the second light source **22** based on the output signal from the second sensor **28**.
[0094] The controller **29** subsequently carries out a wavelength shift correction process for the first light source **21** (step S**13**). Specifically, the controller **29** reads the standard value X.sub.B0 of the X component and the wavelength shift correction coefficient K.sub.BX of the X component out of the control parameters relating to the first light source **21** from the memory. The controller **29** then corrects the standard value X.sub.B0 of the X component based on the first measured value Tm.sub.B and the wavelength shift correction coefficient K.sub.BX of the X component. Specifically, the controller **29** calculates a reference value X.sub.B1 of the X component relating to the first light source **21** by substituting the standard value X.sub.B0 of the X component, the wavelength shift correction coefficient K.sub.BX of the X component, and the first measured value Tm.sub.B into Expression (5) shown above.
[0095] The controller **29** reads the standard value Y.sub.B0 of the Y component and the wavelength shift correction coefficient K.sub.BY of the Y component out of the t control parameters relating to the first light source **21** from the memory. The controller **29** then corrects the standard value Y.sub.B0 of the Y component based on the first measured value Tm.sub.B and the wavelength shift correction coefficient K.sub.BY of the Y component. Specifically, the controller **29** calculates a reference value Y.sub.B1 of the Y component relating to the first light source **21** by substituting the standard value Y.sub.B0 of the Y component, the wavelength shift correction coefficient K.sub.BY of the Y component, and the first measured value Tm.sub.B into Expression (6) shown above.
[0096] The controller **29** further reads the standard value Z.sub.B0 of the Z component and the wavelength shift correction coefficient K.sub.BZ of the Z component out of the control parameters relating to the first light source **21** from the memory. The controller **29** then corrects the standard value Z.sub.B0 of the Z component based on the first measured value Tm.sub.B and the wavelength shift correction coefficient K.sub.BZ of the Z component. Specifically, the controller **29** calculates a reference value Z.sub.B1 of the Z component relating to the first light source **21** by substituting the standard value Z.sub.B0 of the Z component, the wavelength shift correction coefficient K.sub.BZ of the Z component, and the first measured value Tm.sub.B into Expression

(7) shown above.

[0097] The controller **29** subsequently carries out the wavelength shift correction process for the second light source **22** (step S**14**). Specifically, the controller **29** reads the standard value X.sub.Y0 of the X component and the wavelength shift correction coefficient K.sub.YX of the X component out of the control parameters relating to the second light source **22** from the memory. The controller **29** then corrects the standard value X.sub.Y0 of the X component based on the second measured value Tm.sub.Y and the wavelength shift correction coefficient K.sub.YX of the X component. Specifically, the controller **29** calculates a reference value X.sub.Y1 of the X component relating to the second light source **22** by substituting the standard value X.sub.Y0 of the X component, the wavelength shift correction coefficient K.sub.YX of the X component, and the second measured value Tm.sub.Y into Expression (21) below.

[0098] The controller **29** further reads the standard value Y.sub.Y0 of the Y component and the wavelength shift correction coefficient K.sub.YY of the Y component out of the control parameters relating to the second light source **22** from the memory. The controller **29** then corrects the standard value Y.sub.Y0 of the Y component based on the second measured value Tm.sub.Y and the wavelength shift correction coefficient K.sub.YY of the Y component. Specifically, the controller **29** calculates a reference value Y.sub.Y1 of the Y component relating to the second light source **22** by substituting the standard value Y.sub.Y0 of the Y component, the wavelength shift correction coefficient K.sub.YY of the y component, and the second measured value Tm.sub.Y into Expression (22) below.

[0099] The controller **29** still further reads the standard value Z.sub.Y0 of the Z component and the wavelength shift correction coefficient K.sub.YZ of the Z component out of the control parameters relating to the second light source **22** from the memory. The controller **29** then corrects the standard value Z.sub.Y0 of the Z component based on the second measured value Tm.sub.Y and the wavelength shift correction coefficient K.sub.YZ of the Z component. Specifically, the controller **29** calculates a reference value Z.sub.Y1 of the Z component relating to the second light source **22** by substituting the standard value Z.sub.Y0 of the Z component, the wavelength shift correction coefficient K.sub.YZ of the Z component, and the second measured value Tm.sub.Y into Expression (23) below.

[00012]{
$$X_{Y1} = X_{Y0} \text{ .Math. } K_{YX} \text{ .Math. } \mathrm{Tm}_Y \quad (21)$$
$$Y_{Y1} = Y_{Y0} \text{ .Math. } K_{YY} \text{ .Math. } \mathrm{Tm}_Y \quad (22)$$
$$Z_{Y1} = Z_{Y0} \text{ .Math. } K_{YZ} \text{ .Math. } \mathrm{Tm}_Y \quad (23)$$

[0100] The controller **29** subsequently carries out a color matching process (step S**15**). Specifically, the controller **29** calculates a first power setting value Po.sub.B for the first light source **21** and a second power setting value Po.sub.Y for the second light source **22**. A target value of the X component is expressed by Expression (24) below. A target value of the Y component is expressed by Expression (25) below. A target value of the Z component is expressed by Expression (26) below.

[00013]{
$$X = \tfrac{Y}{y}x = \mathrm{Po}_Y \text{ .Math. } X_{Y1} + \mathrm{Po}_B \text{ .Math. } X_{B1} \quad (24)$$
$$Y = Y = \mathrm{Po}_Y \text{ .Math. } Y_{Y1} + \mathrm{Po}_B \text{ .Math. } Y_{B1} \quad (25)$$
$$Z = \tfrac{Y}{y}(1 - x - y) = \mathrm{Po}_Y \text{ .Math. } Z_{Y1} + \mathrm{Po}_B \text{ .Math. } Z_{B1} \quad (26)$$

[0101] When Expressions (24), (25), and (26) described above are transformed into a determinant, Expression (27) below is obtained. The controller **29** solves simultaneous equations based on Expression (27) below to calculate the first power setting value Po.sub.B for the first light source **21** and the second power setting value Po.sub.Y for the second light source **22**.

[00014] $$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} \frac{Y}{y}x \\ Y \\ \frac{Y}{y}(1-x-y) \end{pmatrix} = \begin{pmatrix} 0 & X_{Y1} & X_{B1} \\ 0 & Y_{Y1} & Y_{B1} \\ 0 & Z_{Y1} & Z_{B1} \end{pmatrix} \begin{pmatrix} 0 \\ \mathrm{Po}_Y \\ \mathrm{Po}_B \end{pmatrix} \quad (27)$$

[0102] The controller **29** subsequently carries out an output fixing process for the first light source **21** (step S**16**). Specifically, the controller **29** reads the output fixing coefficients k.sub.B00 to k.sub.Bij out of the control parameters relating to the first light source **21** from the memory. The controller **29** then calculates a first drive current setting value If.sub.B based on the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B. Specifically, the controller **29** calculates the first drive current setting value If.sub.B by substituting the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B into Expression (18) shown above.

[0103] The controller **29** subsequently carries out the output fixing process for the second light source **22** (step S**17**). Specifically, the controller **29** reads the output fixing coefficients k.sub.Y00 to k.sub.Yij out of the control parameters relating to the second light source **22** from the memory. The controller **29** then calculates a second drive current setting value If.sub.Y based on the output fixing coefficients k.sub.Y00 to k.sub.Yij, the second measured value Tm.sub.Y, and the second power setting value Po.sub.Y. Specifically, the controller **29** calculates the second drive current setting value If.sub.Y by substituting the output fixing coefficients k.sub.Y00 to k.sub.Yij, the second measured value Tm.sub.Y, and the second power setting value Po.sub.Y into Expression (28) below.

[00015] $$\mathrm{If}_Y = k_{Y_{00}} \text{ .Math. } \mathrm{Tm}_Y^0 \text{ .Math. } \mathrm{Po}_Y^0 + \text{ .Math. } + k_{Y_{ij}} \text{ .Math. } \mathrm{Tm}_Y^i \text{ .Math. } \mathrm{Po}_Y^j \quad (28)$$

[0104] The controller **29** subsequently controls the drive current supplied to each of the light sources (step S**18**). Specifically, the controller **29** controls the first light source driver **25** in such a way that the value of the first drive current supplied from the first light source driver **25** to the first light source **21** becomes the first drive current setting value If.sub.B. The controller **29** further controls the second light source driver **26** in such a way that the value of the second drive current supplied from the second light source driver **26** to the second light source **22** becomes the second drive current setting value If.sub.Y.

[0105] After the process in step S**18** ends, the controller **29** returns to step S**12** and repeatedly carries out the processes from steps S**12** to S**18** in a predetermined cycle.

Advantages of Second Embodiment

[0106] As described above, the light source apparatus **20** according to the second embodiment includes the first light source **21**, which outputs the first light BL**1**, the second light source **22**, which outputs the second light BL**2**, the first sensor **27**, which detects the first measured value corresponding to the temperature of the first light source **21**, the second sensor **28**, which detects the second measured value corresponding to the temperature of the second light source **22**, and the controller **29**. The controller **29** controls the output from the first light source **21** based on the first measured value, and controls the output from the second light source **22** based on the second measured value.

[0107] According to the light source apparatus **20** described above, since the outputs from the first light source **21** and the second light source **22** are controlled by using two sensors less expensive than the light quantity sensors used in the related art, the cost of the parts of the light source apparatus **20** can be reduced. Since the outputs from the first light source **21** and the second light source **22** are separately controlled based on the temperatures of the first light source **21** and the second light source **22**, respectively, the white light WL output from the light source apparatus **20** can be accurately controlled.

[0108] In the light source apparatus **20** according to the second embodiment, the controller **29** controls the output from the first light source **21** by controlling the first drive current, which is supplied to the first light source **21**, based on the first measured value. The controller **29** further controls the output from the second light source **22** by controlling the second drive current, which is supplied to the second light source **22**, based on the second measured value.

[0109] According to the light source apparatus **20** described above, the output from the first light source **21** can be readily controlled by controlling the first drive current based on the first measured value. The output from the second light source **22** can also be readily controlled by controlling the second drive current based on the second measured value.

[0110] In the light source apparatus **20** according to the second embodiment, the controller **29** calculates the first power setting value Po.sub.B for the first light source **21** and the second power setting value Po.sub.Y Y for the second light source **22** based on the target values (Yxy) expressed in the XYZ color system, the first standard value (X.sub.B0, Y.sub.B0, Z.sub.B0) expressed in the XYZ color system and set in advance in correspondence with the first light source **21**, and the second standard value (X.sub.Y0, Y.sub.Y0, Z.sub.Y0) expressed in the XYZ color system and set in advance in correspondence with the second light source **22**. The controller **29** calculates the first drive current setting value If.sub.B based on the first measured value, the first coefficient (k.sub.B00 to k.sub.Bij) set in advance in correspondence with the first light source **21**, and the first power setting value Po.sub.B. The controller **29** calculates the second drive current setting value If.sub.Y based on the second measured value, the second coefficient (k.sub.Y00 to k.sub.Yij) set in advance in correspondence with the second light source **22**, and the second power setting value Po.sub.Y.

[0111] According to the light source apparatus **20** described above, drive current setting values that can achieve the target values can be accurately calculated in accordance with changes in the temperatures of the light sources.

[0112] In the light source apparatus **20** according to the second embodiment, before calculating the first power setting value Po.sub.B and the second power setting value Po.sub.Y, the controller **29** corrects the first standard value based on the first measured value and the first correction coefficient set in advance in correspondence with the first light source **21**, and corrects the second standard value based on the second measured value and the second correction coefficient set in advance in correspondence with the second light source **22**.

[0113] According to the light source apparatus **20** described above, even when wavelength shifts occur due to changes in the temperatures of the light sources, the first and second standard values are corrected in terms of temperature, so that the drive current setting values can be accurately calculated in accordance with the changes in the temperatures of the light sources.

Third Embodiment of Light Source Apparatus

[0114] FIG. **6** is a block diagram diagrammatically showing the configuration of a light source apparatus **30** according to a third embodiment. The light source apparatus **30** includes a first light source **31**, a beam splitter **32**, a phosphor **33**, a light combiner **34**, a first light source driver **35**, a first sensor **36**, and a controller **37**, as shown in FIG. **6**.

[0115] The first light source **31** outputs first light BL. The first light source **31** outputs blue light as the first light BL by way of example. For example, the first light source **31** may include one or more blue laser diodes to output blue light as the first light BL. The first light source driver **35** supplies the first light source **31** with the first drive current. The output of the first light source **31** is controlled by the first drive current.

[0116] The beam splitter **32** outputs the first light BL to the phosphor **33** and the light combiner **34**. The phosphor **33** converts the first light BL incident from the beam splitter **32** into fourth light YL, which is yellow light. The light combiner **34** combines the blue first light BL incident from the beam splitter **32** and the yellow fourth light YL incident from the phosphor **33** with each other. The light combiner **34** outputs white light WL as a result of the combination of the first light BL and the fourth light YL. That is, the light source apparatus **30** outputs the white light WL.

[0117] The first light source driver **35** supplies the first light source **31** with the first drive current based on a first control signal output from the controller **37**. The value of the first drive current supplied from the first light source driver **35** to the first light source **31** is controlled by the controller **37**.

[0118] The first sensor **36** detects a first measured value corresponding to the temperature of the first light source **31**. The first sensor **36** outputs an electric signal indicating the first measured value to the controller **37**. For example, the first sensor **36** is a thermistor.

[0119] The controller **37** controls the output from the first light source **31**. As will be described later in detail, the controller **37** controls the output from the first light source **31** based on the first measured value detected by the first sensor **36**. More specifically, the controller **37** controls the output from the first light source **31** by controlling the first drive current, which is supplied to the first light source **31**, based on the first measured value.

[0120] For example, the controller **37** includes one or more processors and one or more memories, and control parameters necessary for controlling the first light source **31** are stored in advance in a nonvolatile memory of the controller **37**, as in the controller **15** in the first embodiment. The method for acquiring the control parameters is the same as that in the first embodiment, and is therefore not repeatedly described in the third embodiment.

[0121] As the control parameters necessary for controlling the first light source **31**, output fixing coefficients k.sub.B00 to k.sub.Bij, a standard value X.sub.B0 of the X component, a standard value Y.sub.B0 of the Y component, a standard value Z.sub.B0 of the Z component, a wavelength shift correction coefficient K.sub.BX of the X component, a wavelength shift correction coefficient K.sub.BY of the Y component, and a wavelength shift correction coefficient K.sub.BZ of the Z component are stored in the memory.

[0122] A light source control process carried out by the controller **37** will be described below with reference to FIG. **7**. FIG. **7** is a flowchart showing the light source control process carried out by the controller **37**.

[0123] The controller **37** first sets a target value of the Y component and target values of the color coordinate xy (step S**21**), as shown in FIG. **7**. The target value of the Y component and the target values of the color coordinate xy are examples of a target value expressed in the first color system.

[0124] The controller **37** subsequently measures a value that changes in accordance with the temperature of the first light source **31** (step S**22**). Specifically, the controller **37** acquires a first measured value Tm.sub.B corresponding to the temperature of the first light source **31** based on the output signal from the first sensor **36**.

[0125] The controller **37** subsequently carries out an output fixing process for the first light source **31** (step S**23**). Specifically, the controller **37** reads output fixing coefficients k.sub.B00 to k.sub.Bij out of the control parameters relating to the first light source **31** from the memory. The controller **37** then calculates a first drive current setting value If.sub.B based on the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B. Specifically, the controller **37** calculates the first drive current setting value If.sub.B by substituting the output fixing coefficients k.sub.B00 to k.sub.Bij, the first measured value Tm.sub.B, and the first power setting value Po.sub.B into Expression (18) shown above. Note that when the number of light sources is one as in the third embodiment, the first power setting value Po.sub.B is a fixed value.

[0126] The controller **37** subsequently controls the first drive current supplied to the first light source **31** (step S**24**). Specifically, the controller **37** controls the first light source driver **35** in such a way that the value of the first drive current supplied from the first light source driver **35** to the first light source **31** becomes the first drive current setting value If.sub.B.

[0127] After the process in step S**24** ends, the controller **37** returns to step S**22** and repeatedly carries out the processes from steps S**22** to S**24** in a predetermined cycle.

Advantages of Third Embodiment

[0128] As described above, the light source apparatus **30** according to the third embodiment includes the first light source **31**, which outputs the first light BL, the first sensor **36**, which detects the first measured value corresponding to the temperature of the first light source **31**, and the controller **37**, which controls the output from the first light source **31** based on the first measured value.

[0129] According to the light source apparatus **30** described above, since the output from the first light source **31** is controlled by using one sensor less expensive than the two light quantity sensors used in the related art, the number and the cost of the parts of the light source apparatus **30** can be reduced. Since the output from the first light source **21** is controlled based on the temperature of the first light source **21**, the white light WL output from the light source apparatus **30** can be accurately controlled.

[0130] In the light source apparatus **30** according to the third embodiment, the controller **37** controls the output from the first light source **31** by controlling the first drive current, which is supplied to the first light source **31**, based on the first measured value.

[0131] According to the light source apparatus **30** described above, the output from the first light source **31** can be readily controlled by controlling the first drive current based on the first measured value.

[0132] In the light source apparatus **30** according to the third embodiment, the controller **37** calculates the first drive current setting value If.sub.B based on the first measured value, the first coefficient (k.sub.B00 to k.sub.Bij) set in advance in correspondence with the first light source **31**, and the first power setting value Po.sub.B.

[0133] According to the light source apparatus **30** described above, the first drive current setting value If.sub.B that can achieve the target values can be accurately calculated in accordance with a change in the temperature of the first light source **31**. Furthermore, according to the light source apparatus **30** described above, since the wavelength shift correction process and the color matching process described in the first and second embodiments can be omitted, the calculation load on the controller **37** can be reduced.

Projector

[0134] FIG. **8** diagrammatically shows the configuration of a projector **100** according to the present embodiment.

[0135] The projector **100** is a projection-type image display apparatus that displays video images on a screen SCR, as shown in FIG. **8**. The projector **100** includes the light source apparatus **10** according to the first embodiment, a color separation system **3**, light modulators **4**R, **4**G, and **4**B, a light combining system **5**, and a projection system **6**. Note that the projector **100** may include the light source apparatus **20** according to the second embodiment or the light source apparatus **30** according to the third embodiment in place of the light source apparatus **10** according to the first embodiment.

[0136] The light source apparatus **10** outputs the white light WL toward the color separation system **3**. The color separation system **3** separates the white light WL output from the light source apparatus **10** into red light LR, green light LG, and blue light LB. The color separation system **3** includes a first dichroic mirror **7***a*, a second dichroic mirror **7***b*, a first total reflection mirror **8***a*, a second total reflection mirror **8***b*, a third total reflection mirror **8***c*, a first relay lens **9***a*, and a second relay lens **9***b*.

[0137] The first dichroic mirror **7***a* separates the white light WL from the light source apparatus **10** into the red light LR and light containing the green light LG and the blue light LB. The first dichroic mirror **7***a* transmits the red light LR and reflects the light containing the green light LG and the blue light LB. The second dichroic mirror **7***b* reflects the green light LG and transmits the blue light LB. The second dichroic mirror **7***b* thus separates the light incident from the first dichroic mirror **7***a* into the green light LG and the blue light LB.

[0138] The first total reflection mirror **8***a* is disposed in the optical path of the red light LR, and reflects the red light LR having passed through the first dichroic mirror **7***a* toward the light modulator **4**R. The second total reflection mirror **8***b* and the third total reflection mirror **8***c* are disposed in the optical path of the blue light LB, and guide the blue light LB having passed through the second dichroic mirror **7***b* to the light modulator **4**B. The green light LG is reflected off the second dichroic mirror **7***b* toward the light modulator **4**G.

[0139] The first relay lens **9***a* and the second relay lens **9***b* are disposed in the optical path of the blue light LB on the light exiting side of the second total reflection mirror **8***b*. The first relay lens **9***a* and the second relay lens **9***b* compensate for optical loss of the blue light LB resulting from the fact that the optical path length of the blue light LB is longer than the optical path lengths of the red light LR and the green light LG.

[0140] The light modulator **4**R modulates the red light LR in accordance with image information to form image light corresponding to the red light LR. The light modulator **4**G modulates the green light LG in accordance with image information to form image light corresponding to the green light LG. The light modulator **4**B modulates the blue light LB in accordance with image information to form image light corresponding to the blue light LB.

[0141] The light modulators **4**R, **4**G, and **4**B are each, for example, a transmissive liquid crystal panel. Polarizers that are not shown are disposed at the light incident and exiting sides of each of the liquid crystal panels.

[0142] A field lens **2**R is disposed on the light incident side of the light modulator **4**R. The field lens **2**R parallelizes the red light LR to be incident on the light modulator **4**R. A field lens **2**G is disposed on the light incident side of the light modulator **4**G. The field lens **2**G parallelizes the green light LG to be incident on the light modulator **4**G. A field lens **2**B is disposed on the light incident side of the light modulator **4**B. The field lens **2**B parallelizes the blue light LB to be incident on the light modulator **4**B.

[0143] The image light output from the light modulator **4**R, the image light output from the light modulator **4**G, and the image light output from the light modulator **4**B enter the light combining system **5**. The light combining system **5** combines the image light corresponding to the red light LR, the image light corresponding to the green light LG, and the image light corresponding to the blue light LB with one another and outputs the combined image light toward the projection system **6**. The light combining system **5** is, for example, a cross dichroic prism.

[0144] The projection system **6** includes multiple projection lenses. The projection system **6** enlarges the combined image light from the light combining system **5** and projects the enlarged image light toward the screen SCR. Enlarged video images are thus displayed on the screen SCR.

[0145] Note that the technical scope of the present disclosure is not limited to the embodiments described above, and various changes can be made thereto without departing from the intent of the present disclosure.

[0146] For example, the projector **100** including the light source apparatus **10** is presented by way of example in the embodiments described above, and the light source apparatus according to the present disclosure can be widely used as a light source apparatus of an electronic instrument other than the projector **100**.

SUMMARY OF PRESENT DISCLOSURE

[0147] The present disclosure will be summarized below as additional remarks.

[0148] (Additional Remark 1) A light source apparatus including: a first light source configured to output first light; a first sensor configured to detect a first measured value corresponding to a temperature of the first light source; and a controller configured to control an output from the first light source based on the first measured value.

[0149] The light source apparatus according to Additional Remark 1, in which the output from the first light source is controlled by using one sensor less expensive than the two light quantity sensors used in the related art, allows reduction in the number and the cost of the parts of the light source apparatus. Since the output from the first light source is controlled based on the temperature of the first light source, the light output from the light source apparatus can be accurately

controlled.
[0150] (Additional Remark 2) The light source apparatus according to Additional Remark 1, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value.
[0151] The light source apparatus according to Additional Remark 2 can readily control the output from the first light source by controlling the first drive current based on the first measured value.
[0152] (Additional Remark 3) The light source apparatus according to Additional Remark 2, wherein the controller is configured to calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and a first power setting value for the first light source.
[0153] In the light source apparatus according to Additional Remark 3, the value that sets a first drive current that can achieve a target value can be accurately calculated in accordance with a change in the temperature of the first light source.
[0154] (Additional Remark 4) The light source apparatus according to Additional Remark 1 further including: a second light source configured to output second light; and a second sensor configured to detect a second measured value corresponding to a temperature of the second light source, wherein the controller is configured to control an output from the second light source based on the second measured value.
[0155] The light source apparatus according to Additional Remark 4, in which the outputs from the first and second light sources are controlled by using two sensors less expensive than the light quantity sensors used in the related art, allows reduction in the cost of the parts of the light source apparatus. Since the outputs from the first and second light sources are separately controlled based on the temperatures of the first and second light sources, respectively, the light output from the light source apparatus can be accurately controlled.
[0156] (Additional Remark 5) The light source apparatus according to Additional Remark 4, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value, and control the output from the second light source by controlling a second drive current, which is supplied to the second light source, based on the second measured value.
[0157] The light source apparatus according to Additional Remark 5 can readily control the output from the first light source by controlling the first drive current based on the first measured value. The output from the second light source can also be readily controlled by controlling the second drive current based on the second measured value.
[0158] (Additional Remark 6) The light source apparatus according to Additional Remark 5, wherein the controller is configured to calculate a first power setting value for the first light source and a second power setting value for the second light source based on a target value expressed in a first color system, a first standard value expressed in the first color system and set in advance in correspondence with the first light source, and a second standard value expressed in the first color system and set in advance in correspondence with the second light source, calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and the first power setting value, and calculate a value that sets the second drive current based on the second measured value, a second coefficient set in advance in correspondence with the second light source, and the second power setting value.
[0159] In the light source apparatus according to Additional Remark 6, the values that set drive currents that can achieve the target value can be accurately calculated in accordance with changes in the temperatures of the light sources.
[0160] (Additional Remark 7) The light source apparatus according to Additional Remark 6, wherein before calculating the first and second power setting values, the controller is configured to correct the first standard value based on the first measured value and a first correction coefficient set in advance in correspondence with the first light source, and correct the second standard value

based on the second measured value and a second correction coefficient set in advance in correspondence with the second light source.

[0161] In the light source apparatus according to Additional Remark 7, even when wavelength shifts occur due to changes in the temperatures of the light sources, the first and second standard values are corrected in terms of temperature, so that the values that set the drive currents can be accurately calculated in accordance with the changes in the temperatures of the light sources.

[0162] (Additional Remark 8) The light source apparatus according to Additional Remark 4 further including: a third light source configured to output third light; and a third sensor configured to detect a third measured value corresponding to a temperature of the third light source, wherein the controller is configured to control an output from the third light source based on the third measured value.

[0163] The light source apparatus according to Additional Remark 8, in which the outputs from the first, second, and third light sources are controlled by using three sensors less expensive than the light quantity sensors used in the related art, allows reduction in the cost of the parts of the light source apparatus. Since the outputs from the first, second, and third light sources are separately controlled based on the temperatures of the first, second, and third light sources, respectively, the light output from the light source apparatus can be accurately controlled.

[0164] (Additional Remark 9) The light source apparatus according to Additional Remark 8, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value, control the output from the second light source by controlling a second drive current, which is supplied to the second light source, based on the second measured value, and control the output from the third light source by controlling a third drive current, which is supplied to the third light source, based on the third measured value.

[0165] The light source apparatus according to Additional Remark 9 can readily control the output from the first light source by controlling the first drive current based on the first measured value. The output from the second light source can also be readily controlled by controlling the second drive current based on the second measured value. The output from the third light source can still further be readily controlled by controlling the third drive current based on the third measured value.

[0166] (Additional Remark 10) The light source apparatus according to Additional Remark 9, wherein the controller is configured to calculate a first power setting value for the first light source, a second power setting value for the second light source, and a third power setting value for the third light source based on a target value expressed in a first color system, a first standard value expressed in the first color system and set in advance in correspondence with the first light source, a second standard value expressed in the first color system and set in advance in correspondence with the second light source, and a third standard value expressed in the first color system and set in advance in correspondence with the third light source, calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and the first power setting value, calculate a value that sets the second drive current based on the second measured value, a second coefficient set in advance in correspondence with the second light source, and the second power setting value, and calculate a value that sets the third drive current based on the third measured value, a third coefficient set in advance in correspondence with the third light source, and the third power setting value.

[0167] In the light source apparatus according to Additional Remark 10, the values that set drive currents that can achieve the target value can be accurately calculated in accordance with changes in the temperatures of the light sources.

[0168] (Additional Remark 11) The light source apparatus according to Additional Remark 10, wherein before calculating the first, second, and third power setting values, the controller is configured to correct the first standard value based on the first measured value and a first correction coefficient set in advance in correspondence with the first light source, correct the second standard value based on the second measured value and a second correction coefficient set in advance in correspondence with the second light source, and correct the third standard value based on the third measured value and a third correction coefficient set in advance in correspondence with the third light source.

[0169] In the light source apparatus according to Additional Remark 11, even when wavelength shifts occur due to changes in the temperatures of the light sources, the first, second, and third standard values are corrected in terms of temperature, so that the values that set the drive currents can be accurately calculated in accordance with the changes in the temperatures of the light sources.

[0170] (Additional Remark 12) A projector including: the light source apparatus according to any one of Additional Remarks 1 to 11; a light modulator configured to modulate light output from the light source apparatus; and a projection system configured to project the light modulated by the light modulator.

[0171] The projector according to Additional Remark 12, which includes the light source apparatus that allows reduction in the number and the cost of the parts of the light source apparatus, allows reduction in the number and the cost of the parts of the projector itself.

Claims

1. A light source apparatus comprising: a first light source configured to output first light; a first sensor configured to detect a first measured value corresponding to a temperature of the first light source; and a controller configured to control an output from the first light source based on the first measured value.

2. The light source apparatus according to claim 1, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value.

3. The light source apparatus according to claim 2, wherein the controller is configured to calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and a first power setting value for the first light source.

4. The light source apparatus according to claim 1 further comprising: a second light source configured to output second light; and a second sensor configured to detect a second measured value corresponding to a temperature of the second light source, wherein the controller is configured to control an output from the second light source based on the second measured value.

5. The light source apparatus according to claim 4, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value, and control the output from the second light source by controlling a second drive current, which is supplied to the second light source, based on the second measured value.

6. The light source apparatus according to claim 5, wherein the controller is configured to calculate a first power setting value for the first light source and a second power setting value for the second light source based on a target value expressed in a first color system, a first standard value expressed in the first color system and set in advance in correspondence with the first light source, and a second standard value expressed in the first color system and set in advance in correspondence with the second light source, calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and the first power setting value, and calculate a value that sets the second drive current based on the second measured value, a second coefficient set in advance in correspondence with the second light source, and the second power setting value.

7. The light source apparatus according to claim 6, wherein before calculating the first and second power setting values, the controller is configured to correct the first standard value based on the first measured value and a first correction coefficient set in advance in correspondence with the first light source, and correct the second standard value based on the second measured value and a second correction coefficient set in advance in correspondence with the second light source.
8. The light source apparatus according to claim 4 further comprising: a third light source configured to output third light; and a third sensor configured to detect a third measured value corresponding to a temperature of the third light source, wherein the controller is configured to control an output from the third light source based on the third measured value.
9. The light source apparatus according to claim 8, wherein the controller is configured to control the output from the first light source by controlling a first drive current, which is supplied to the first light source, based on the first measured value, control the output from the second light source by controlling a second drive current, which is supplied to the second light source, based on the second measured value, and control the output from the third light source by controlling a third drive current, which is supplied to the third light source, based on the third measured value.
10. The light source apparatus according to claim 9, wherein the controller is configured to calculate a first power setting value for the first light source, a second power setting value for the second light source, and a third power setting value for the third light source based on a target value expressed in a first color system, a first standard value expressed in the first color system and set in advance in correspondence with the first light source, a second standard value expressed in the first color system and set in advance in correspondence with the second light source, and a third standard value expressed in the first color system and set in advance in correspondence with the third light source, calculate a value that sets the first drive current based on the first measured value, a first coefficient set in advance in correspondence with the first light source, and the first power setting value, calculate a value that sets the second drive current based on the second measured value, a second coefficient set in advance in correspondence with the second light source, and the second power setting value, and calculate a value that sets the third drive current based on the third measured value, a third coefficient set in advance in correspondence with the third light source, and the third power setting value.
11. The light source apparatus according to claim 10, wherein before calculating the first, second, and third power setting values, the controller is configured to correct the first standard value based on the first measured value and a first correction coefficient set in advance in correspondence with the first light source, correct the second standard value based on the second measured value and a second correction coefficient set in advance in correspondence with the second light source, and correct the third standard value based on the third measured value and a third correction coefficient set in advance in correspondence with the third light source.
12. A projector comprising: the light source apparatus according to claim 1; a light modulator configured to modulate light output from the light source apparatus; and a projection system configured to project the light modulated by the light modulator.